BURLINGTON PIERS, ON, Canada

WMO: **714370**

Lat: **43.3005N** Lon: **79.7916W** Elev: **254** StdP: **14.56** Time Zone: **-5.00 (NAE)** Period: **94-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	3.3	8.1	-7.0	3.8	5.2	-2.6	4.8	9.7	27.8	24.5	25.3	25.1	9.9	0	0.515	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	12.5	86.5	70.5	83.6	69.3	80.8	68.3	73.8	81.2	72.2	79.4	70.8	77.9	9.3	230	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	71.5	117.6	78.0	69.8	110.7	76.1	68.3	104.9	75.3	37.6	81.1	36.0	79.8	34.9	77.9	84.6	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	24.1	20.6	17.9	DB	-1.2	90.8	5.8	2.7	-5.4	92.7	-8.8	94.3	-12.1	95.9	-16.3	97.8	(4)
(5)				WB	-2.2	76.6	5.5	2.6	-6.1	78.5	-9.3	80.1	-12.4	81.5	-16.3	83.4	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours		DBAvg	49.1	26.2	27.3	34.4	44.6	55.3	65.8	71.5	70.8	64.6	52.9	42.1	32.7	(6)
(7)			DBStd	17.82	10.49	9.39	8.85	7.18	7.63	7.09	5.52	4.90	6.43	7.30	7.82	8.40	(7)
(8)			HDD50	2929	739	637	487	195	26	0	0	0	0	51	257	537	(8)
(9)			HDD65	6385	1203	1057	948	614	319	76	8	7	83	381	688	1001	(9)
(10)			CDD50	2611	0	0	5	31	190	474	667	645	439	141	18	1	(10)
(11)			CDD65	593	0	0	0	0	19	100	209	187	72	6	0	0	(11)
(12)			CDH74	3147	0	0	0	4	134	586	1251	853	297	22	0	0	(12)
(13)			CDH80	754	0	0	0	0	22	157	339	175	59	2	0	0	(13)
(14)	Wind		WSAvg	8.3	9.6	9.3	9.3	9.0	7.4	6.9	6.6	6.9	7.8	8.7	9.0	9.4	(14)
(15)	Precipitation		PrecAvg	34.8	2.8	2.4	2.5	3.0	3.1	3.1	3.2	3.0	3.0	2.9	2.9	2.7	(15)
(16)			PrecMax	43.4	5.8	4.5	5.5	5.7	5.7	7.7	7.4	6.3	6.8	5.9	5.2	5.4	(16)
(17)			PrecMin	27.0	1.2	1.3	1.1	1.2	1.0	0.9	0.7	1.2	1.0	0.7	0.8	1.4	(17)
(18)			PrecStd	4.3	1.1	0.9	1.2	1.2	1.2	1.6	1.7	1.2	1.3	1.3	1.1	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		55.0	54.8	64.3	73.1	83.0	88.4	90.6	88.9	86.0	76.8	65.0	56.7	(19)
(20)			MCWB		51.0	48.7	53.7	58.0	65.6	70.6	72.8	71.4	71.2	64.0	56.4	52.0	(20)
(21)		2%	DB		48.2	47.7	56.7	66.5	78.0	84.8	86.9	84.4	80.9	71.6	60.8	51.4	(21)
(22)			MCWB		44.4	42.1	48.8	54.8	63.6	69.2	70.6	70.7	68.4	61.7	53.9	47.3	(22)
(23)		5%	DB		43.1	41.8	50.9	60.5	72.8	80.6	83.8	81.5	77.1	67.5	57.2	46.3	(23)
(24)			MCWB		39.6	37.3	44.8	50.8	61.1	67.5	69.3	69.0	66.9	59.7	50.7	42.2	(24)
(25)		10%	DB		39.0	38.4	44.9	55.0	67.9	76.8	80.8	78.9	74.1	63.5	53.1	42.8	(25)
(26)			MCWB		36.3	35.1	40.0	46.7	58.1	65.8	68.4	68.2	66.3	57.3	47.9	39.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		51.6	50.3	56.6	60.3	69.2	74.2	77.1	75.5	73.5	68.3	58.8	52.5	(27)
(28)			MCDB		54.8	53.8	63.5	68.2	78.9	82.8	85.1	82.4	80.5	74.0	63.2	56.0	(28)
(29)		2%	WB		45.1	42.8	50.3	56.6	65.9	71.6	74.3	73.2	71.3	63.4	55.2	47.7	(29)
(30)			MCDB		47.5	47.2	54.8	64.8	74.7	80.1	82.0	80.3	77.4	68.8	59.9	50.9	(30)
(31)		5%	WB		39.6	38.0	44.9	51.6	62.6	69.6	72.4	71.7	69.5	60.5	51.2	42.9	(31)
(32)			MCDB		42.1	41.1	50.2	59.6	70.9	77.5	79.9	78.5	75.2	65.5	56.0	45.6	(32)
(33)		10%	WB		36.5	35.4	40.2	47.6	59.2	67.5	70.8	70.3	67.5	58.3	48.3	39.8	(33)
(34)			MCDB		38.8	38.0	44.3	54.4	66.5	75.4	78.1	76.7	72.7	63.3	52.6	42.5	(34)
(35)	Mean Daily Temperature Range		MDBR		10.9	11.3	11.1	12.3	13.3	12.5	12.5	11.1	11.6	10.9	9.8	9.0	(35)
(36)		5% DB	MCDBR		14.6	15.4	18.7	23.0	21.6	18.3	16.5	15.1	15.9	17.0	16.0	13.1	(36)
(37)			MCWBR		13.5	13.1	13.8	14.6	11.7	9.0	7.7	7.6	8.4	10.9	12.4	11.5	(37)
(38)		5% WB	MCDBR		13.9	14.5	17.9	21.7	20.8	16.4	14.4	12.4	12.8	14.7	14.9	13.1	(38)
(39)			MCWBR		13.5	13.1	14.3	15.7	13.5	9.6	8.2	7.9	8.5	10.9	12.4	12.2	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.296	0.335	0.338	0.386	0.409	0.446	0.437	0.422	0.389	0.366	0.315	0.299	(40)
(41)		taud			2.403	2.264	2.388	2.289	2.278	2.201	2.249	2.280	2.353	2.396	2.507	2.450	(41)
(42)		Ebn at Noon			266	270	286	279	274	264	264	265	265	257	258	254	(42)
(43)		Edh at Noon			24	33	33	40	42	45	43	40	34	29	22	22	(43)
(44)	All-Sky Solar Radiation	RadAvg			446	685	1079	1406	1711	1872	1872	1647	1316	795	497	344	(44)
(45)		RadStd			44	68	115	153	176	132	136	85	123	73	66	39	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page